

PLANNING

4.1 Implementation Roadmap & Sprint Milestones

The development schedule for this project component was organized into clear sprints to manage system dependencies, validate configurations, and minimize risk before deploying to production instance clusters.

Sprint 1: Schema Setup & Variable Mapping

Phase I

Focuses on creating the Service Catalog item variables, testing foundational user data strings, and ensuring manager reference variables populate correctly in sandbox scopes.

Sprint 2: Approval Flow Orchestration

Phase II

Focuses on designing the primary flow trigger inside Workflow Studio, implementing the *Ask For Approval* system action, and mapping dynamic data pills across records.

Sprint 3: Conditional Branching & Layout Customization

Phase III

Focuses on constructing the evaluation logic forks, building the fallback update record steps, and embedding structural HTML styling elements within the email engine.

Sprint 4: Edge-Case Verification & Quality Testing

Phase IV

Focuses on simulating real-world scenarios across approval states, analyzing transactional system tracking logs, and confirming user input data renders without formatting issues.

4.2 Risk Management and Fallback Protocols

Defensive technical planning ensures the automated flow can handle incomplete or malformed database fields safely without throwing system errors.

Identified Risk: Missing Manager Fields

If an active employee profile lacks an assigned manager value, the workflow engine encounters a null value, causing the approval block to fault and stall mid-execution.

System Resolution Strategy

A pre-flight validation check has been engineered directly into the intake logic. If the supervisor lookup field returns empty, the flow intercepts the error, bypasses the standard path, raises an automated alert to system administrators, and safely redirects approval tasks to a fallback IT management triage group.

4.3 Project Schedule Milestone Targets

The project timeline tracks key deliverables along the approval engine's path to deployment. Each step must pass formal system validation gates before work proceeds to subsequent tasks.

Milestone Identifier	Target Phase	Engineering Focus	Duration	Pre-requisite Dependencies
MS-APP-01	Catalog Item Variable	Data Mapping	2 Business Days	Requirement Sign-off
MS-APP-02	Flow Trigger	Lookup Actions Verified	3 Business Days	MS-APP-01 Complete
MS-APP-03	HTML Rejection Output	Engine Active	3 Business Days	MS-APP-02 Complete
MS-APP-04	UAT System	Sign-off & Code Verification	4 Business Days	MS-APP-03 Complete

4.4 Functional Testing Framework Specification

The testing framework runs verification checks across two operational boundaries to confirm processing logic performs exactly as expected:

- **Unit Path Validation:**

Confirms that submitting a catalog request correctly wakes up the process trigger, pulls user record values dynamically from the database, and builds corresponding approval entries without field dropping or data truncation.

- **Integration Fork Validation:**

Confirms the flow processes conditional loops correctly. Simulating a positive approval response must advance the workflow to the procurement queue, while submitting a rejection must immediately update the record state to *Closed Incomplete* and route structural comments directly into outbound HTML layouts.